

V35

September 16, 2026

height of track

```
[5]: import numpy as np
import matplotlib.pyplot as plt
```

```
[2]: h = np.array([11.1, 12.8, 11.5, 12.8, 12.7])
err_h = 0.05

print(np.mean(h))
error_mean = np.std(h, ddof=1)/np.sqrt(len(h))
print(error_mean)
print(np.sqrt(error_mean**2 + err_h**2))
```

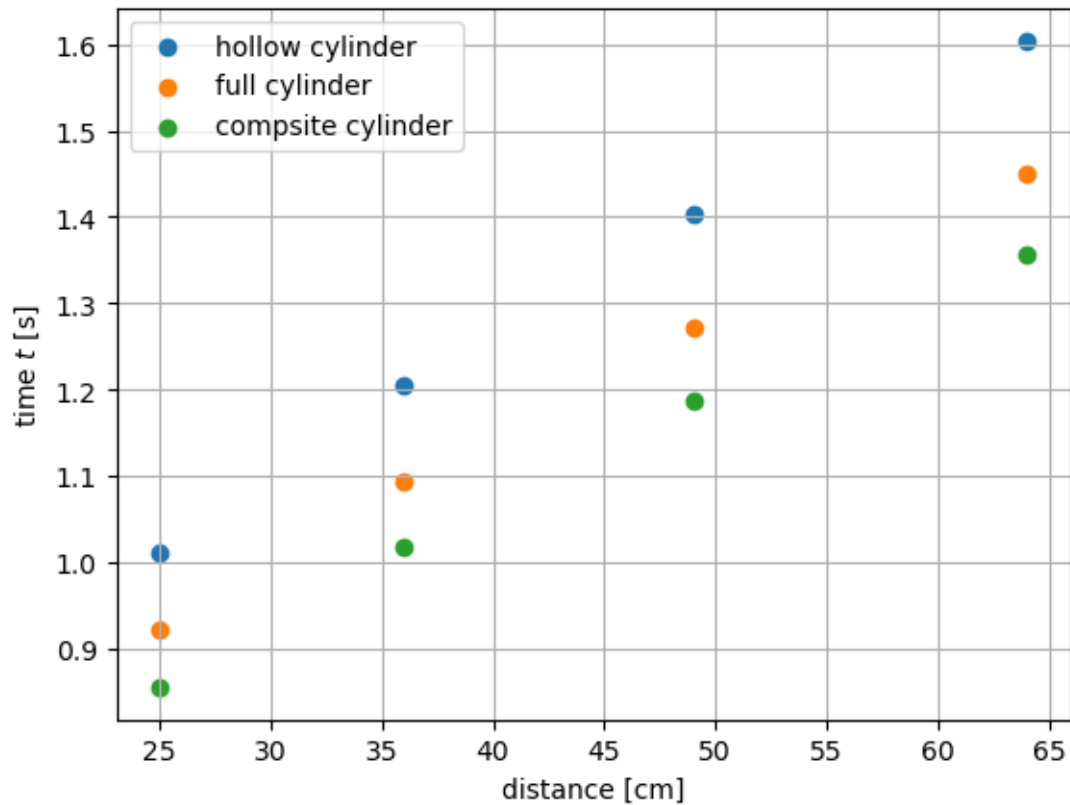
```
12.180000000000001
0.3652396473549936
0.3686461718233353
```

```
[ ]: t_hollow = np.array([1.011, 1.204, 1.403, 1.604])
t_full = np.array([0.921, 1.094, 1.272, 1.451])
t_compose = np.array([0.855, 1.018, 1.186, 1.356])

x = np.array([25, 36, 49, 64])

plt.scatter(x, t_hollow, label="hollow cylinder")
plt.scatter(x, t_full, label="full cylinder")
plt.scatter(x, t_compose, label="composite cylinder")
plt.xlabel("distance [cm]")
plt.ylabel("time $t$ [s]")
plt.grid(True)
plt.legend()
```

```
[ ]: <matplotlib.legend.Legend at 0x11bb9cb90>
```



Quantitative A3:

```
[13]: x = np.array([25, 36, 49, 64])

t_hollow = np.array([
    [1.019, 1.212, 1.410, 1.610],
    [1.027, 1.223, 1.426, 1.629],
    [1.014, 1.209, 1.410, 1.613],
    [1.022, 1.215, 1.415, 1.614],
    [1.009, 1.201, 1.400, 1.600]
])
t_full = np.array([
    [0.922, 1.096, 1.278, 1.459],
    [0.893, 1.068, 1.245, 1.425],
    [0.900, 1.075, 1.255, 1.435],
    [0.901, 1.074, 1.254, 1.435],
    [0.913, 1.087, 1.266, 1.446]
])

mean_hollow = np.mean(t_hollow, axis=0)
```

```

std_hollow = np.std(t_hollow, axis=0, ddof=1)/np.sqrt(t_hollow.shape[0]) # 
    ↳Standard error of the mean
mean_full = np.mean(t_full, axis=0)
std_full = np.std(t_full, axis=0, ddof=1)/np.sqrt(t_full.shape[0]) # Standard 
    ↳error of the mean

print("Hollow cylinder:")
for distance, mean, std in zip(x, mean_hollow, std_hollow):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")

print("\nFull cylinder:")
for distance, mean, std in zip(x, mean_full, std_full):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")

```

Hollow cylinder:

```

25 cm: mean = 1.0182 s, std = 0.0031 s
36 cm: mean = 1.2120 s, std = 0.0036 s
49 cm: mean = 1.4122 s, std = 0.0042 s
64 cm: mean = 1.6132 s, std = 0.0047 s

```

Full cylinder:

```

25 cm: mean = 0.9058 s, std = 0.0052 s
36 cm: mean = 1.0800 s, std = 0.0050 s
49 cm: mean = 1.2596 s, std = 0.0057 s
64 cm: mean = 1.4400 s, std = 0.0058 s

```

a4

```

[20]: # messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])

m_diff_h = np.array([])
for i in [m1h, m2h, m3h, m4h, m5h]:
    m_diff_h = np.append(m_diff_h, i[1] - i[0])

```

```

m_diff_f = np.array([])
for i in [m1f, m2f, m3f, m4f, m5f]:
    m_diff_f = np.append(m_diff_f, i[1] - i[0])

err_t = 0.001
print("Hollow cylinder time differences:")
print(m_diff_h)
print("Mean time difference for hollow cylinder:", np.mean(m_diff_h))
print("Standard error for hollow cylinder:", np.std(m_diff_h, ddof=1)/
      ↪len(m_diff_h))
print('Full error for hollow cylinder:', np.sqrt((np.std(m_diff_h, ddof=1)/
      ↪len(m_diff_h))**2 + err_t**2))

print("\nFull cylinder time differences:")

print(m_diff_f)
print("Mean time difference for full cylinder:", np.mean(m_diff_f))
print("Standard error for full cylinder:", np.std(m_diff_f, ddof=1)/
      ↪len(m_diff_f))
print('Full error for full cylinder:', np.sqrt((np.std(m_diff_f, ddof=1)/
      ↪len(m_diff_f))**2 + err_t**2))

```

```

Hollow cylinder time differences:
[0.27  0.271 0.27  0.27  0.269]
Mean time difference for hollow cylinder: 0.27000000000000001
Standard error for hollow cylinder: 0.00014142135623730964
Full error for hollow cylinder: 0.0010099504938362077

Full cylinder time differences:
[0.243 0.244 0.243 0.244 0.243]
Mean time difference for full cylinder: 0.2434
Standard error for full cylinder: 0.00010954451150102927
Full error for full cylinder: 0.001005982107196743

```